# Chapter 9. Epilogue—Life Outside $\mathbb{R}^n$

Differentiable manifolds and Riemannian manifolds

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## Overview

Manifolds often occur as abstract spaces; we sketch how to move outside $\mathbb{R}^n$.

## §41. Differentiable Manifolds and Riemannian Manifolds

**Motivation.** Abstract manifolds keep only coordinate charts; forms and Stokes' theorem carry over once tangent vectors are intrinsic.

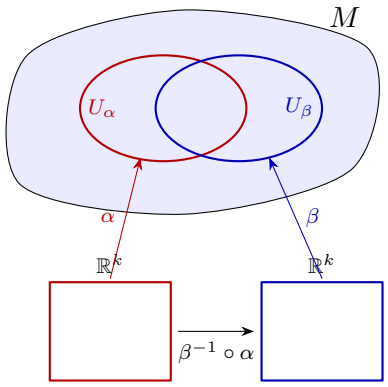



**Figure.** *Coordinate patches $\alpha$, $\beta$ of an abstract manifold overlap; the transition map $\beta^{-1}\circ\alpha$ between open sets of $\mathbb{R}^k$ must be $C^r$. Smoothness on $M$ is defined through these charts, without any ambient $\mathbb{R}^n$.*

### 41.1 Differentiable manifolds

**Definition 41.1 (differentiable manifold)**

Let $M$ be a metric space. Suppose there is a collection of homeomorphisms $\alpha_i : U_i \to V_i$, where $U_i$ is open in $\mathbb{H}^k$ or $\mathbb{R}^k$ and $V_i$ is open in $M$, such that the $V_i$ cover $M$. ($\alpha_i$ is a **homeomorphism** if it carries $U_i$ onto $V_i$ one-to-one and both $\alpha_i$, $\alpha_i^{-1}$ are continuous.) Suppose the $\alpha_i$ overlap with class $C^\infty$: the **transition function** $\alpha_i^{-1}\circ\alpha_j$ is of class $C^\infty$ whenever $V_i \cap V_j$ is non-empty. The $\alpha_i$ are **coordinate patches** on $M$, as is any other homeomorphism $\alpha : U \to V$ ($U$ open in $\mathbb{H}^k$ or $\mathbb{R}^k$, $V$ open in $M$) that overlaps the $\alpha_i$ with class $C^\infty$. $M$ with this collection of patches is a **differentiable $k$-manifold** (of class $C^\infty$).

For $k = 1$ we again allow domains open in $\mathbb{L}^1$. If there is a patch $\alpha : U \to V$ about $p$ with $U$ open in $\mathbb{R}^k$, $p$ is an **interior point**; otherwise a **boundary point**; the boundary points form $\partial M$. If $\alpha : U \to V$ is a patch about $p$, then $p \in \partial M$ iff $U$ is open in $\mathbb{H}^k$ and $p = \alpha(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^{k-1} \times 0$ (proof as in Lemma 24.2). Throughout, $M$ denotes a differentiable $k$-manifold.

**Definition 41.2 (orientation)**

Patches $\alpha_0, \alpha_1$ on $M$ **overlap positively** if $\det D(\alpha_1^{-1}\circ\alpha_0) > 0$. If $M$ can be covered by patches overlapping positively, $M$ is **orientable**. An **orientation** of $M$ consists of such a covering together with all other patches overlapping these positively; an **oriented manifold** is a manifold with an orientation. If $\{\alpha_i\}$ is an orientation, $\{\alpha_i \circ r\}$ ($r$ the reflection of $\mathbb{R}^k$) is the **opposite** orientation.

If $\partial M \neq \emptyset$, then $\partial M$ is a differentiable $(k-1)$-manifold without boundary, with patches $\alpha \circ b$ ($\alpha$ a patch about $p \in \partial M$, $b(x_1, \ldots, x_{k-1}) = (x_1, \ldots, x_{k-1}, 0)$), as in Theorem 24.3. If $\alpha_0, \alpha_1$ overlap positively, so do $\alpha_0 \circ b$ and $\alpha_1 \circ b$ (Theorem 34.1). So if $M$ is oriented, $\partial M$ can be oriented by

restricting the patches of the orientation of $M$; for $k$ even this is the **induced orientation** of $\partial M$, for $k$ odd its opposite is.

**Definition 41.3 ($C^\infty$ maps between manifolds)**

Let $M$, $N$ be differentiable manifolds of dimensions $k$, $n$, $A \subset M$ and $f : A \to N$. $f$ is **of class** $C^\infty$ if for each $x \in A$ there are a patch $\alpha : U \to V$ on $M$ about $x$ and a patch $\beta : W \to Y$ on $N$ about $y = f(x)$ such that $\beta^{-1} \circ f \circ \alpha$ is of class $C^\infty$ as a map of a subset of $\mathbb{R}^k$ into $\mathbb{R}^n$. (Independent of the patches, since transition functions are $C^\infty$.) A one-to-one $f : M \to N$ onto $N$ with $f$ and $f^{-1}$ of class $C^\infty$ is a **diffeomorphism**.

## 41.2 Tangent vectors

For $M$ in $\mathbb{R}^n$, a tangent vector at $p$ is the velocity $(p; D\gamma(t_0))$ of a $C^\infty$ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$. For abstract $M$ we cannot differentiate $\gamma$, but for a patch $\alpha : U \to V$ about $p$, $\alpha^{-1} \circ \gamma$ maps a subset of $\mathbb{R}^1$ into $\mathbb{R}^k$, so we may think of the "derivative" of $\gamma$ at $t_0$ as the function $\mathbf{v}$ assigning to each patch $\alpha$ about $p$ the matrix $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$. If $\alpha_0, \alpha_1$ are two patches about $p$, the chain rule gives

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \qquad g = \alpha_1^{-1} \circ \alpha_0, \quad \mathbf{x}_0 = \alpha_0^{-1}(p).$$

**Definition 41.4 (tangent vector, tangent space)**

Given $p \in M$, a **tangent vector** to $M$ at $p$ is a function $\mathbf{v}$ assigning to each coordinate patch $\alpha : U \to V$ about $p$ a $k$ by 1 column matrix $\mathbf{v}(\alpha)$, such that for any two patches $\alpha_0$, $\alpha_1$ about $p$,

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \tag{$*$}$$

where $g = \alpha_1^{-1} \circ \alpha_0$ and $\mathbf{x}_0 = \alpha_0^{-1}(p)$. The entries of $\mathbf{v}(\alpha)$ are the **components** of $\mathbf{v}$ with respect to $\alpha$.

By $(*)$, $\mathbf{v}$ is determined by its components in a single patch, and we may define $(a\mathbf{v} + b\mathbf{w})(\alpha) = a\mathbf{v}(\alpha) + b\mathbf{w}(\alpha)$. The tangent vectors at $p$ form the **tangent space** $\mathcal{T}_p(M)$, a $k$-dimensional vector space: if $\alpha(\mathbf{x}) = p$, the map $\mathbf{v} \mapsto (\mathbf{x}; \mathbf{v}(\alpha))$ is a linear isomorphism of $\mathcal{T}_p(M)$ onto $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k)$, whose inverse is denoted $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$; so $\alpha_*(\mathbf{x}; \mathbf{v}(\alpha)) = \mathbf{v}$. For a $C^\infty$ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$, the **velocity vector** is defined by $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$; every tangent vector at $p$ is the velocity vector of some curve.

**Tangent vectors as derivations**

To $\mathbf{v} \in \mathcal{T}_p(M)$ associate the operator $X_{\mathbf{v}}$ on $C^\infty$ functions near $p$, the **derivative with respect to v**. If $\mathbf{v}$ is the velocity of $\gamma$ at $t_0$ and $\alpha(\mathbf{x}) = p$, then from $f \circ \gamma = (f \circ \alpha) \circ (\alpha^{-1} \circ \gamma)$,

$$\frac{d(f \circ \gamma)}{dt}(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot D(\alpha^{-1} \circ \gamma)(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha),$$

which depends only on $f$ and $\mathbf{v}$. So define $X_{\mathbf{v}}(f) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha)$; it is independent of $\alpha$, $X_{\mathbf{v}+\mathbf{w}} = X_{\mathbf{v}} + X_{\mathbf{w}}$, $X_{c\mathbf{v}} = cX_{\mathbf{v}}$, and for $M = \mathbb{R}^k$ it is the directional derivative of $f$ with respect to $\mathbf{v}$. It satisfies (1) *locality*: if $f, g$ agree near $p$, $X_{\mathbf{v}}(f) = X_{\mathbf{v}}(g)$;
(2) *linearity*: $X_{\mathbf{v}}(af + bg) = aX_{\mathbf{v}}(f) + bX_{\mathbf{v}}(g)$;
(3) the *product rule*: $X_{\mathbf{v}}(f \cdot g) = X_{\mathbf{v}}(f)g(p) + f(p)X_{\mathbf{v}}(g)$. Every operator satisfying (1)–(3) equals $X_{\mathbf{v}}$ for a unique $\mathbf{v} \in \mathcal{T}_p(M)$, so tangent vectors may be defined intrinsically as such operators.

## 41.3 Forms on $M$

**Definition 41.5 (forms on a manifold)**

An $l$-**form** on $M$ is a function $\omega$ assigning to each $p \in M$ an alternating $l$-tensor $\omega(p) \in \mathcal{A}^l(\mathcal{T}_p(M))$. If $\alpha : U \to V$ is a patch about $p$ with $\alpha(\mathbf{x}) = p$, let $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$ and $T^* : \mathcal{A}^l(\mathcal{T}_p(M)) \to \mathcal{A}^l(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$; define $(\alpha^*\omega)(\mathbf{x}) = T^*(\omega(p))$. $\omega$ is **of class** $C^\infty$ **near** $p$ if $\alpha^*\omega$ is of class $C^\infty$ near $\mathbf{x}$ (independent of $\alpha$), and of class $C^\infty$ if this holds for every patch.

Henceforth all forms are $C^\infty$. $\Omega^l(M)$ is the space of $l$-forms on $M$. There are no elementary forms on $M$ giving a canonical form, but $\alpha^*\omega = \sum_{[I]} f_I\, dx_I$ in $\mathbb{R}^k$; the $f_I$ are the **components** of $\omega$ with respect to $\alpha$.

**Definition 41.6 (differential)**

For an $l$-form $\omega$ on $M$, $d\omega$ is defined as follows: given $p \in M$ and tangent vectors $\mathbf{v}_1, \dots, \mathbf{v}_{l+1}$ at $p$, choose a patch $\alpha$ about $p$ with $\alpha(\mathbf{x}) = p$ and set

$$d\omega(p)(\mathbf{v}_1, \dots, \mathbf{v}_{l+1}) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{v}_1(\alpha)), \dots, (\mathbf{x}; \mathbf{v}_{l+1}(\alpha))).$$

That is, pull $\omega$ back to $\alpha^*\omega$ in $\mathbb{R}^k$, pull the $\mathbf{v}_i$ back to tangent vectors in $\mathbb{R}^k$, and apply $d$ in $\mathbb{R}^k$. Equivalently (with $\mathbf{a}_i = \mathbf{v}_i(\alpha)$), $d\omega(p)(\alpha_*(\mathbf{x}; \mathbf{a}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{a}_{l+1})) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{a}_1), \dots, (\mathbf{x}; \mathbf{a}_{l+1}))$, i.e. $d\omega$ is the unique $(l+1)$-form on $M$ such that

$$\alpha^*(d\omega) = d(\alpha^*\omega)$$

for every coordinate patch $\alpha$ on $M$. (The $d$ on the right is the operator in $\mathbb{R}^k$.)

This is independent of the patch, and $d\omega$ is $C^\infty$.

**Theorem 41.1**

Let $M$ be a compact differentiable manifold. Given a covering of $M$ by coordinate patches, there exist functions $\phi_i : M \to \mathbb{R}$ of class $C^\infty$, for $i = 1, \dots, l$, such that:
(1) $\phi_i(p) \geq 0$ for each $p \in M$.
(2) For each $i$, the set Support $\phi_i$ is covered by one of the given coordinate patches.
(3) $\sum \phi_i(p) = 1$ for each $p \in M$.

**Proof.** Given $p \in M$, choose a patch $\alpha : U \to V$ of the covering about $p$, $\alpha(\mathbf{x}) = p$, and a non-negative $C^\infty$ function $f : U \to \mathbb{R}$ whose support is compact and contained in $U$, with $f(\mathbf{x}) > 0$. Define $\psi_p : M \to \mathbb{R}$ by $\psi_p(y) = f(\alpha^{-1}(y))$ for $y \in V$ and $\psi_p(y) = 0$ otherwise. Since $f \circ \alpha^{-1}$ vanishes outside a compact subset of $V$, $\psi_p$ is $C^\infty$ on $M$. $\psi_p$ is positive on an open set $U_p$ about $p$. Cover $M$ by finitely many $U_p$, for $p = p_1, \dots, p_l$, and set $\lambda = \sum_{j=1}^{l} \psi_{p_j}$ and $\phi_i = (1/\lambda)\psi_{p_i}$. ∎

**Definition 41.7 (integral of a form)**

Let $M$ be a compact oriented differentiable $k$-manifold and $\omega$ a $k$-form on $M$. If Support $\omega$ lies in a single patch $\alpha : U \to V$ of the orientation, $\int_M \omega = \int_{\mathrm{Int}\, U} \alpha^*\omega$. In general, with $\phi_1, \dots, \phi_l$ as in Theorem 41.1,

$$\int_M \omega = \sum_{i=1}^{l} \Big[ \int_M \phi_i \omega \Big].$$

The usual argument shows this is well-defined and linear.

**Theorem 41.2 (Stokes' theorem)**

Let $M$ be a compact, oriented differentiable $k$-manifold. Let $\omega$ be a $k-1$ form on $M$. If $\partial M$ is non-empty, give $\partial M$ the induced orientation; then

$$\int_M d\omega = \int_{\partial M} \omega.$$

If $\partial M$ is empty, then $\int_M d\omega = 0$.

**Proof.** The proof of Theorem 37.2 goes through verbatim: all computations were carried out within coordinate patches, so no changes are needed. The special conventions for $k = 1$ and $\partial M$ a 0-manifold are handled as before (Theorem 37.4). ■

Chapter 8 also generalizes: the **deRham group** $H^k(M)$ is the quotient of the closed $k$-forms on $M$ by the exact ones, and there are methods to compute it, including a general *Mayer–Vietoris theorem* relating the deRham groups of $M = U \cup V$, $U$, $V$ and $U \cap V$ (see Bott–Tu). $H^k(M)$ is a diffeomorphism invariant and even a *topological* one (homeomorphic $M, N$ have isomorphic $H^k$), by *deRham's theorem*: closed modulo exact forms is isomorphic to the real cohomology algebra of algebraic topology.

## 41.4 Riemannian manifolds

$\mathrm{v}(M)$ and $\int_M f\,dV$ do not make sense on an abstract manifold. By §36, the volume of a compact oriented $k$-manifold $M$ in $\mathbb{R}^n$ is $\mathrm{v}(M) = \int_M \omega_V$, where the volume form $\omega_V$ has value 1 on *any orthonormal basis* of $\mathcal{T}_p(M)$ in its natural orientation. This uses the inner product $\mathcal{T}_p(M) \subset p \times \mathbb{R}^n$ inherits from $\mathbb{R}^n$; an abstract $\mathcal{T}_p(M)$ needs one supplied.

**Definition 41.8 (Riemannian metric)**

A **Riemannian metric** on a differentiable $k$-manifold $M$ is an inner product $\langle \mathbf{v}, \mathbf{w} \rangle$ on each tangent space $\mathcal{T}_p(M)$, required to be of class $C^\infty$ as a 2-tensor field on $M$. A **Riemannian manifold** is a differentiable manifold together with a Riemannian metric.

Every differentiable manifold has a Riemannian metric (the proof uses a partition of unity), but it is not unique. A Riemannian metric gives a volume function $V(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ on $k$-tuples in $\mathcal{T}_p(M)$ (as in the exercises of §21), so one defines, for continuous $f$ supported in a single patch $\alpha : U \to V$,

$$\int_M f\,dV = \int_{\mathrm{Int}\,U} (f \circ \alpha)\, V(\alpha_*(\mathbf{x}; \mathbf{e}_1), \ldots, \alpha_*(\mathbf{x}; \mathbf{e}_k)),$$

in general via a partition of unity as in §25, and $\mathrm{v}(M) = \int_M dV$. If $M$ is compact, oriented and Riemannian, $\int_M \omega = \int_M \lambda\,dV$ where $\lambda(p)$ is the value of $\omega(p)$ on an orthonormal $k$-tuple in $\mathcal{T}_p(M)$ belonging to its natural orientation; if $\lambda \equiv 1$, $\omega$ is the **volume form** $\omega_V$ of $M$, and $\mathrm{v}(M) = \int_M \omega_V$.

The *length* of a smooth curve $\gamma : [a, b] \to M$ is $\int_{t=a}^{t=b} \|\gamma_*(t; \mathbf{e}_1)\|$, the integral of the norm of its velocity vector (using the inner product on $\mathcal{T}_p(M)$); one studies *geodesics* (curves of minimal length joining two points), *curvature*, and so on—the subject of *Riemannian geometry*.

**What does not generalize**

Manifolds, forms and Stokes' theorem generalize; the vector-field form of Stokes (§38) does not: the translation functions of §31, grad, div (in $\mathbb{R}^n$), curl (in $\mathbb{R}^3$) and normal vectors all depend on the ambient space.

## Selected exercises

**Motivation.** Exercises (§41): proofs of the facts stated above, each in coordinate patches.

**Exercise 41.1 (tangent vectors are velocities)**

Show that if $\mathbf{v} \in \mathcal{T}_p(M)$, then $\mathbf{v}$ is the velocity vector of some $C^\infty$ curve $\gamma$ in $M$ passing through $p$.

**Solution.**

*Step 1.* Choose a patch $\alpha : U \to V$ about $p$, $\alpha(\mathbf{x}) = p$, and let $\mathbf{a} = \mathbf{v}(\alpha)$. Since $U$ is open in $\mathbb{R}^k$ or in $\mathbb{H}^k$, there is $r > 0$ such that every point of $\mathbb{H}^k$ within distance $r$ of $\mathbf{x}$ lies in $U$ (every point of $\mathbb{R}^k$, if $U$ is open in $\mathbb{R}^k$). Let $\epsilon = r/(1 + \|\mathbf{a}\|)$.

*Step 2.* Choose the parameter interval $J$; in each case $\|t\mathbf{a}\| \le \epsilon\|\mathbf{a}\| < r$ for $t \in J$. If $U$ is open in $\mathbb{R}^k$, let $J = [-\epsilon, \epsilon]$; then $\mathbf{x} + t\mathbf{a} \in U$. If $U$ is open in $\mathbb{H}^k$, it suffices that the last coordinate $x_k + ta_k$ be $\ge 0$ on $J$:

if $x_k > 0$, take $J = [-\epsilon', \epsilon']$ with $\epsilon' = \min\{\epsilon,\, x_k/(1 + |a_k|)\}$, so that $x_k + ta_k > x_k - x_k = 0$;

if $x_k = 0$, take $J = [-\epsilon, \epsilon]$, $[0, \epsilon]$ or $[-\epsilon, 0]$ according as $a_k = 0$, $a_k > 0$ or $a_k < 0$.

*Step 3.* Define $\gamma : J \to M$ by $\gamma(t) = \alpha(\mathbf{x} + t\mathbf{a})$. Then $\gamma(0) = p$, and $\alpha^{-1} \circ \gamma(t) = \mathbf{x} + t\mathbf{a}$ is $C^\infty$, so $\gamma$ is a $C^\infty$ curve. Its velocity vector $\mathbf{w}$ at $t_0 = 0$ has

$$\mathbf{w}(\alpha) = D(\alpha^{-1} \circ \gamma)(0) = \mathbf{a} = \mathbf{v}(\alpha)$$

(a one-sided derivative if 0 is an endpoint of $J$). By condition $(*)$ of Definition 41.4, a tangent vector is determined by its components in one patch, so $\mathbf{w} = \mathbf{v}$. (For $k = 1$ and a patch on $\mathbb{L}^1$ the same argument applies with the inequalities reversed.) ■

**Exercise 41.3 ($d\omega$ is well-defined)**

If $\omega$ is an $l$-form on $M$, show that $d\omega$ in Definition 41.6 is independent of the choice of the coordinate patch $\alpha$.

**Solution.** Let $\alpha_0 : U_0 \to V_0$ and $\alpha_1 : U_1 \to V_1$ be patches about $p$ with $\alpha_0(\mathbf{x}_0) = p = \alpha_1(\mathbf{x}_1)$. Let $W = \alpha_0^{-1}(V_0 \cap V_1)$ and $g = \alpha_1^{-1} \circ \alpha_0 : W \to U_1$; $g$ is $C^\infty$ and $g(\mathbf{x}_0) = \mathbf{x}_1$.

*Step 1.* ($\alpha_{0*} = \alpha_{1*} \circ g_*$.) Let $\mathbf{x} \in W$, $\mathbf{y} = g(\mathbf{x})$, $\mathbf{a} \in \mathbb{R}^k$, and let $\mathbf{v} = \alpha_{0*}(\mathbf{x}; \mathbf{a})$, i.e. $\mathbf{v}(\alpha_0) = \mathbf{a}$. By $(*)$ of Definition 41.4, $\mathbf{v}(\alpha_1) = Dg(\mathbf{x}) \cdot \mathbf{a}$, so

$$\alpha_{0*}(\mathbf{x}; \mathbf{a}) = \alpha_{1*}(\mathbf{y}; Dg(\mathbf{x}) \cdot \mathbf{a}) = \alpha_{1*}(g_*(\mathbf{x}; \mathbf{a})).$$

*Step 2.* ($\alpha_0^*\omega = g^*(\alpha_1^*\omega)$ on $W$.) For tangent vectors $(\mathbf{x}; \mathbf{a}_1), \dots, (\mathbf{x}; \mathbf{a}_l)$, Definition 41.5 and Step 1 give

$$(\alpha_0^*\omega)(\mathbf{x})\big((\mathbf{x}; \mathbf{a}_i)\big)_i = \omega(p')\big(\alpha_{1*}(g_*(\mathbf{x}; \mathbf{a}_i))\big)_i = (\alpha_1^*\omega)(\mathbf{y})\big(g_*(\mathbf{x}; \mathbf{a}_i)\big)_i,$$

with $p' = \alpha_0(\mathbf{x})$; the right side is $(g^*(\alpha_1^*\omega))(\mathbf{x})((\mathbf{x}; \mathbf{a}_i))_i$ by Definition 32.1.

*Step 3.* ($d$ commutes with $g^*$.) On the interior of $W$, Theorem 32.3 and Step 2 give $d(\alpha_0^*\omega) = d(g^*(\alpha_1^*\omega)) = g^*(d(\alpha_1^*\omega))$. Both sides are continuous on $W$, and every point of $W$ is a limit of interior points of $W$ (if $W$ is open in $\mathbb{H}^k$, points of $W \cap (\mathbb{R}^{k-1} \times 0)$ are limits of points of $W$ with $x_k > 0$); so the identity holds on all of $W$.

*Step 4.* (Conclusion.) Let $\mathbf{v}_1, \dots, \mathbf{v}_{l+1} \in \mathcal{T}_p(M)$, $\mathbf{a}_i = \mathbf{v}_i(\alpha_0)$ and $\mathbf{b}_i = \mathbf{v}_i(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{a}_i$. By Step 3 at $\mathbf{x}_0$,

$$d(\alpha_0^*\omega)(\mathbf{x}_0)\big((\mathbf{x}_0; \mathbf{a}_i)\big)_i = d(\alpha_1^*\omega)(\mathbf{x}_1)\big(g_*(\mathbf{x}_0; \mathbf{a}_i)\big)_i$$

$$= d(\alpha_1^*\omega)(\mathbf{x}_1)\big((\mathbf{x}_1;\mathbf{b}_i)\big)_i.$$

The two sides are the values of $d\omega(p)(\mathbf{v}_1,\dots,\mathbf{v}_{l+1})$ computed with $\alpha_0$ and with $\alpha_1$, so they agree. ■

**Exercise 41.5 (Riemannian metrics exist)**

Show that any compact differentiable manifold $M$ has a Riemannian metric.

**Solution.** Here an inner product $\langle\cdot,\cdot\rangle_p$ on each $\mathcal{T}_p(M)$ is *of class* $C^\infty$ if for every patch $\alpha: U \to V$ the functions $G_{rs}(\mathbf{x}) = \langle \alpha_*(\mathbf{x};\mathbf{e}_r), \alpha_*(\mathbf{x};\mathbf{e}_s)\rangle_{\alpha(\mathbf{x})}$ are $C^\infty$ on $U$.

*Step 1.* (Local metrics.) Cover $M$ by patches and let $\phi_1,\dots,\phi_l$ be as in Theorem 41.1, with Support $\phi_i$ covered by the patch $\alpha_i: U_i \to V_i$. For $p \in V_i$ and $\mathbf{v},\mathbf{w} \in \mathcal{T}_p(M)$ let $\langle \mathbf{v},\mathbf{w}\rangle_i = \mathbf{v}(\alpha_i)\cdot\mathbf{w}(\alpha_i)$ (dot product of components). Since $\mathbf{v} \mapsto \mathbf{v}(\alpha_i)$ is a linear isomorphism onto $\mathbb{R}^k$, this is an inner product on $\mathcal{T}_p(M)$.

*Step 2.* (Gluing.) Define

$$\langle \mathbf{v},\mathbf{w}\rangle_p = \sum_{i=1}^{l} \phi_i(p)\langle \mathbf{v},\mathbf{w}\rangle_i,$$

where a term with $\phi_i(p) = 0$ is read as 0 (so only patches with $p \in \text{Support}\,\phi_i \subset V_i$ contribute). Each term is symmetric and bilinear, hence so is the sum. Since $\phi_i \geq 0$ and $\sum_i \phi_i(p) = 1$, some $\phi_j(p) > 0$; then for $\mathbf{v} \neq \mathbf{0}$,

$$\langle \mathbf{v},\mathbf{v}\rangle_p \geq \phi_j(p)\|\mathbf{v}(\alpha_j)\|^2 > 0.$$

So $\langle\cdot,\cdot\rangle_p$ is an inner product on $\mathcal{T}_p(M)$.

*Step 3.* (Smoothness.) Let $\alpha: U \to V$ be a patch and $g_i = \alpha_i^{-1}\circ\alpha$ on $U_i' = \alpha^{-1}(V \cap V_i)$. By $(*)$ of Definition 41.4, $\alpha_*(\mathbf{x};\mathbf{e}_r)(\alpha_i) = Dg_i(\mathbf{x})\cdot\mathbf{e}_r$, so the $i$th term of $G_{rs}$ is

$$h_i(\mathbf{x}) = \phi_i(\alpha(\mathbf{x}))\,(Dg_i(\mathbf{x})\mathbf{e}_r)\cdot(Dg_i(\mathbf{x})\mathbf{e}_s) \quad (\mathbf{x}\in U_i'), \qquad h_i(\mathbf{x}) = 0 \quad \text{otherwise.}$$

$h_i$ is $C^\infty$ on the open set $U_i'$ ($\phi_i\circ\alpha$ and $g_i$ are $C^\infty$), and $h_i = 0$ on the open set $\alpha^{-1}(V - \text{Support}\,\phi_i)$; these two open sets cover $U$ because Support $\phi_i \subset V_i$. Hence $h_i$ and $G_{rs} = \sum_i h_i$ are $C^\infty$ on $U$. ■

**Exercise 41.6 (tangent vectors as derivations)**

Let $p$ be a point of $M - \partial M$. Let $X$ be an operator on $C^\infty$ real-valued functions defined near $p$ satisfying locality, linearity and the product rule. Show there is exactly one $\mathbf{v} \in \mathcal{T}_p(M)$ with $X = X_{\mathbf{v}}$, as follows.
(a) Let $F$ be $C^\infty$ on the open cube $U = \{\mathbf{x} : |\mathbf{x}| < \epsilon\}$ in $\mathbb{R}^k$. Show there are $C^\infty$ functions $g_1,\dots,g_k$ on $U$ with $F(\mathbf{x}) - F(\mathbf{0}) = \sum_j x_j g_j(\mathbf{x})$.
(b) Show that $D_jF(\mathbf{0}) = g_j(\mathbf{0})$.
(c) Show that $X(c) = 0$ if $c$ is a constant function.
(d) Show that there is at most one $\mathbf{v}$ with $X = X_{\mathbf{v}}$.
(e) Show that there is a $\mathbf{v}$ with $X = X_{\mathbf{v}}$.

**Solution.**

(a) Let

$$g_j(\mathbf{x}) = \int_{u=0}^{u=1} D_jF(x_1,\dots,x_{j-1},ux_j,0,\dots,0).$$

*Step 1.* The point $(x_1,\dots,x_{j-1},ux_j,0,\dots,0)$ lies in $U$ for $\mathbf{x}\in U$, $u\in[0,1]$, and the integrand is a $C^\infty$ function of $(\mathbf{x},u)$. Applying Leibniz's rule (Theorem 39.1) on $Q\times[0,1]$ for closed cubes $Q\subset U$, repeatedly, every partial derivative of $g_j$ of every order exists and is continuous; so $g_j$ is $C^\infty$.

*Step 2.* Substituting $t = ux_j$ (both sides are 0 if $x_j = 0$) and using the fundamental theorem of calculus,

$$\begin{aligned} x_j g_j(\mathbf{x}) &= \int_{t=0}^{t=x_j} D_j F(x_1, \dots, x_{j-1}, t, 0, \dots, 0) \\ &= F(x_1, \dots, x_j, 0, \dots, 0) - F(x_1, \dots, x_{j-1}, 0, \dots, 0). \end{aligned}$$

*Step 3.* Summing over $j = 1, \dots, k$, the right sides telescope to $F(\mathbf{x}) - F(\mathbf{0})$.

(b) At $\mathbf{x} = \mathbf{0}$ the integrand is the constant $D_j F(\mathbf{0})$, so $g_j(\mathbf{0}) = D_j F(\mathbf{0})$.

(c) By the product rule, $X(1) = X(1 \cdot 1) = X(1) \cdot 1 + 1 \cdot X(1) = 2X(1)$, so $X(1) = 0$; by linearity $X(c) = cX(1) = 0$.

(d) Let $\alpha : U \to V$ be a patch with $U$ open in $\mathbb{R}^k$ and $\alpha(\mathbf{0}) = p$ (translate the domain if necessary), and let $h = \alpha^{-1} : V \to U$ with components $h_1, \dots, h_k$. Each $h_i$ is $C^\infty$ near $p$, since $h_i \circ \alpha(\mathbf{x}) = x_i$. If $X = X_{\mathbf{v}}$, then

$$X(h_i) = D(h_i \circ \alpha)(\mathbf{0}) \cdot \mathbf{v}(\alpha) = \mathbf{e}_i^{\mathsf{T}} \mathbf{v}(\alpha) = v_i(\alpha).$$

So the components of $\mathbf{v}(\alpha)$ are the numbers $X(h_i)$, and $\mathbf{v}$ is determined by $X$.

(e)

*Step 1.* With $\alpha$, $h$ as in (d), let $\mathbf{v}$ be the tangent vector at $p$ with $\mathbf{v}(\alpha) = (X(h_1), \dots, X(h_k))$. Let $f$ be $C^\infty$ near $p$ and $F = f \circ \alpha$, which is $C^\infty$ near $\mathbf{0}$. Choose $\epsilon > 0$ so that the cube $U_\epsilon = \{|\mathbf{x}| < \epsilon\}$ lies in $U$ and $\alpha(U_\epsilon)$ lies in the domain of $f$. By (a), $F(\mathbf{x}) = \sum_j x_j g_j(\mathbf{x}) + F(\mathbf{0})$ on $U_\epsilon$.

*Step 2.* Since $\alpha$ is a homeomorphism onto the open set $V$, $\alpha(U_\epsilon)$ is a neighborhood of $p$, and on it

$$f = F \circ h = \sum_j h_j \cdot (g_j \circ h) + F(\mathbf{0}),$$

where $g_j \circ h$ is $C^\infty$ near $p$ because $(g_j \circ h) \circ \alpha = g_j$.

*Step 3.* By locality, linearity, the product rule and (c),

$$X(f) = \sum_j \big[ X(h_j)\, g_j(h(p)) + h_j(p)\, X(g_j \circ h) \big] + X(F(\mathbf{0})).$$

Here $h(p) = \mathbf{0}$, so $h_j(p) = 0$, and $g_j(\mathbf{0}) = D_j F(\mathbf{0})$ by (b). Hence

$$X(f) = \sum_j D_j F(\mathbf{0})\, X(h_j) = D(f \circ \alpha)(\mathbf{0}) \cdot \mathbf{v}(\alpha) = X_{\mathbf{v}}(f).$$

■